Adaptive Staircase Experiment
PSY310 - Lab in Psychology
Lab Report

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GitHub Link:

Introduction

Adaptive staircase procedures are an efficient method to measure sensory thresholds and are commonly used in psychophysical research. They involve an algorithm that actively adjusts the intensity of the stimuli according to the performance of the participant.[1] Unlike other methods which have complex stimulus placement rules and threshold estimates, the simple and flexible adaptive staircase method uses participants' previous responses to determine the next trial placement, making it easier or more difficult.[2] Historically, this method is most used to test visual perception such as colour and brightness perception; and recently it has been applied to proprioception by researchers of motor control.[3]

[1] Hairston & Maldjian, 2008
[2] Leek, 2001
[3] Hoseini et al., 2015

Method

Participant

The participant is an 18 year old male undergraduate student at Ahmedabad University. Full consent was provided and he was informed of the trial's objective, to determine the direction of the grating's tilt.

Materials and Procedure

This experiment was created on a desktop computer with a resolution of 1920x1200 pixels using PsychoPy-2025.1.1. The procedure assessed the participant's ability to evaluate the direction of the grating's tilt. 100 trials of the experiment were conducted.

The trial began with a cross-shaped fixation presented for 500 milliseconds followed by the stimuli, a sinusoidal grating with a Gaussian mask (Figure 1), presented for 300 milliseconds. A code was added so that the grating was randomly tilted either to the left or right. The maximum intensity of the tilt was 10 degrees and the minimum was 1 degree, which was adjusted with the step sizes of 2, 1 and 0.5 degrees based on whether the participant answered the previous trial correctly or incorrectly, as shown in Figure 2. The participant was instructed to press the right arrow key if the grating had a rightward tilt, and the left arrow key if it had a leftward tilt. The responses were collected on PsychoPy in an Excel spreadsheet, which were then analysed to determine the accuracy.

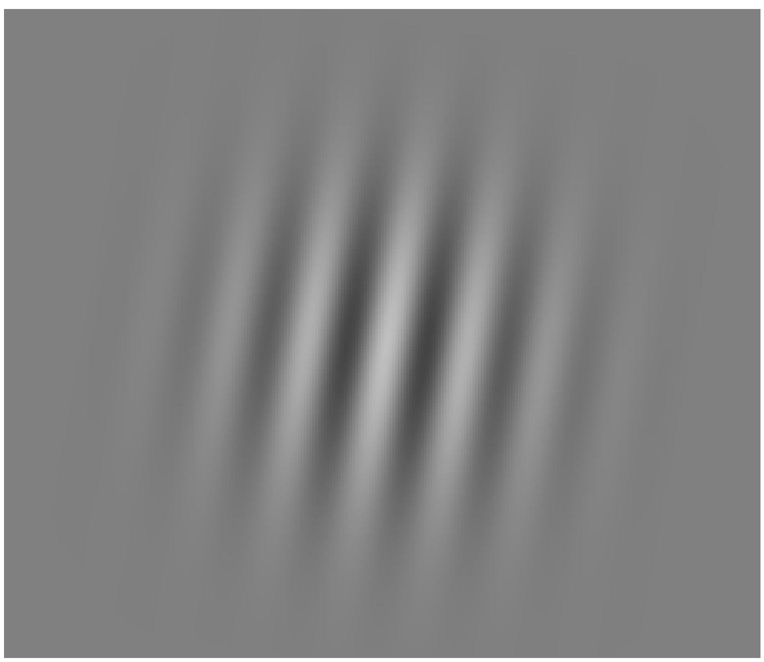

Figure 1 - Sinusoidal grating with a Gaussian mask, tilted 10 degrees rightward.

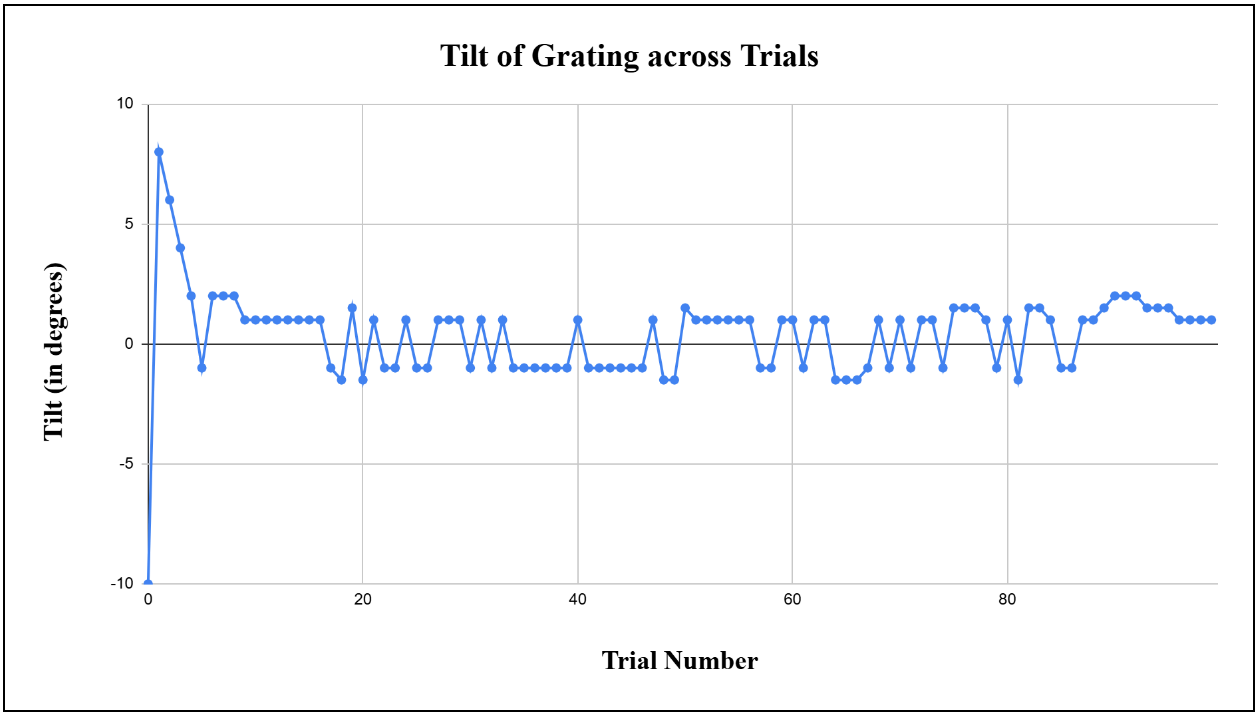

Figure 2 - The changes in tilt intensity from the first to the hundredth trial.

Results

The intensities of the last 5 reversal points are: 1.5, 1, 1.5, 1 and 2. The average of these points is 1.4, which is the point of absolute threshold. It indicates that beyond this intensity, the participant can no longer identify the direction of the grating's tilt. The participant responses with a 92% accuracy over the 100 trials, as shown in Figure 3.

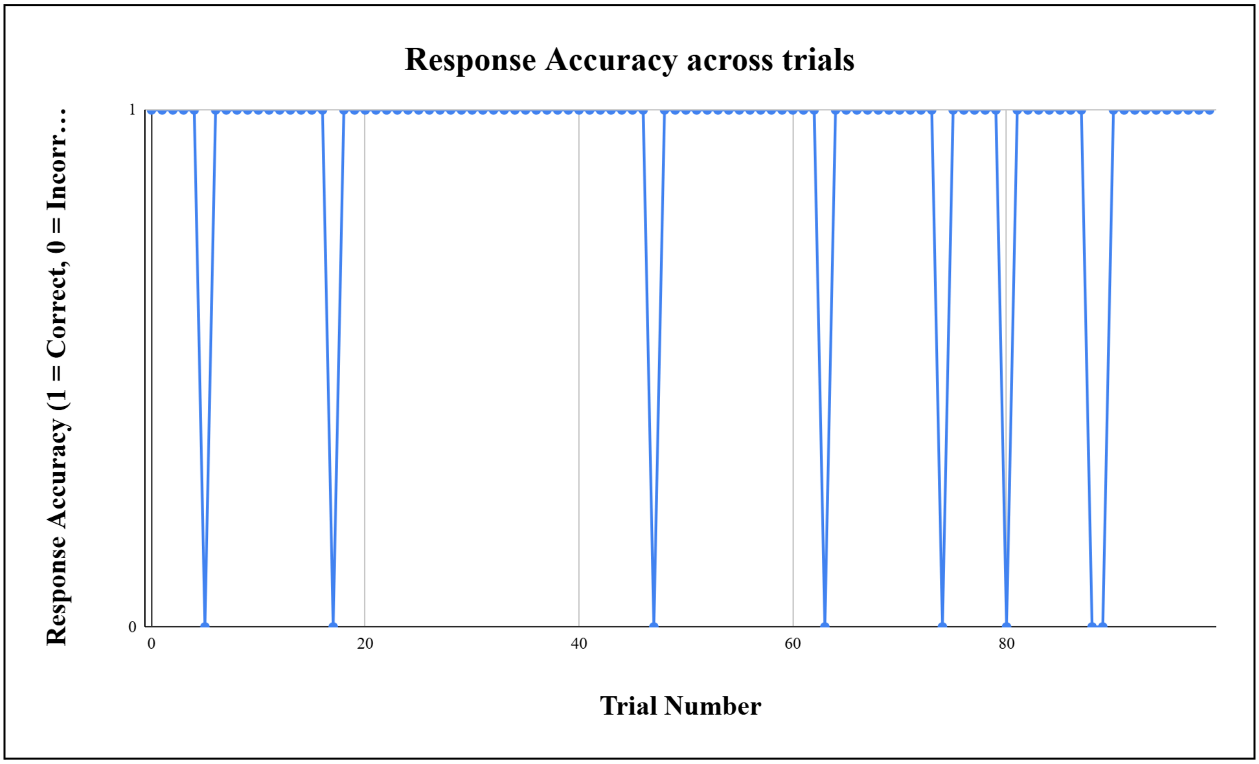

Figure 3 - Participant's accuracy over the 100 trials.

Discussion

After analysing the participant's responses, a 92% accuracy was observed out of the 100%, and a threshold value of 1.4 degrees was obtained. This indicates that the lowest level of tilt that the participant was able to differentiate between a left or right tilt was 1.4 degrees, which is a great level of accuracy as the minimum tilt in the experiment was 1 degree.

However, the ecological validity of this experiment is low, as these results would not be applicable in external environments outside the lab. Also, as there was only 1 participant, it is not possible to generalise any insight from the results. Thus, further research is required.

References

Hairston, W. D., & Maldjian, J. A. (2008). An adaptive staircase procedure for the E-Prime programming environment. *Computer Methods and Programs in Biomedicine*.

Hoseini, N., Sexton, B. M., Kurtz, K., Liu, Y., & Block, H. J. (2015, August 14). *Adaptive Staircase Measurement of Hand Proprioception | PLOS One*. PLOS; DOI. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0135757

Leek, M.R. Adaptive procedures in psychophysical research. *Perception & Psychophysics* 63, 1279–1292 (2001). https://doi.org/10.3758/BF03194543